

Taylor and Maclaurin Series



The Taylor series formula

- 1 State the general Taylor series formula for $f(x)$ centered at $x = a$.
 - 2 Find the Taylor series for $f(x) = \ln x$ centered at $a = 1$, up through the $(x - 1)^3$ term.
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Common Maclaurin series

- 3 State the Maclaurin series for e^x , $\sin x$, and $\cos x$.
 - 4 Use the Maclaurin series for e^x to find the Maclaurin series for e^{-x^2} .
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Using known series to build new series

- 5 Use the Maclaurin series for $\sin x$ to find the first three nonzero terms of the Maclaurin series for $x \sin x$.
 - 6 Use the known Taylor series for $\cos x$ to estimate $\cos(0.1)$ using the first three nonzero terms.
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The Lagrange error bound

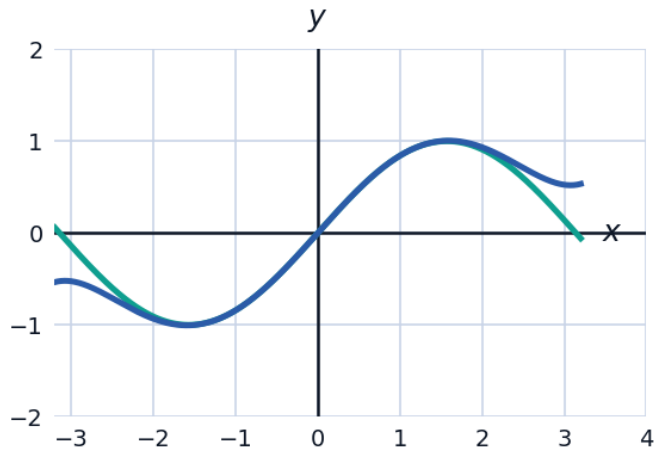
- 7 State the Lagrange Error Bound formula for the remainder $R_n(x)$ of a Taylor polynomial of degree n .
 - 8 Use the Lagrange Error Bound to estimate the maximum error when approximating $\sin(0.5)$ with the Taylor polynomial $P_3(x) = x - \frac{x^3}{6}$ (degree 3, centered at 0), given $|f^{(4)}(z)| \leq 1$ for all z .
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Radius of convergence of Taylor series

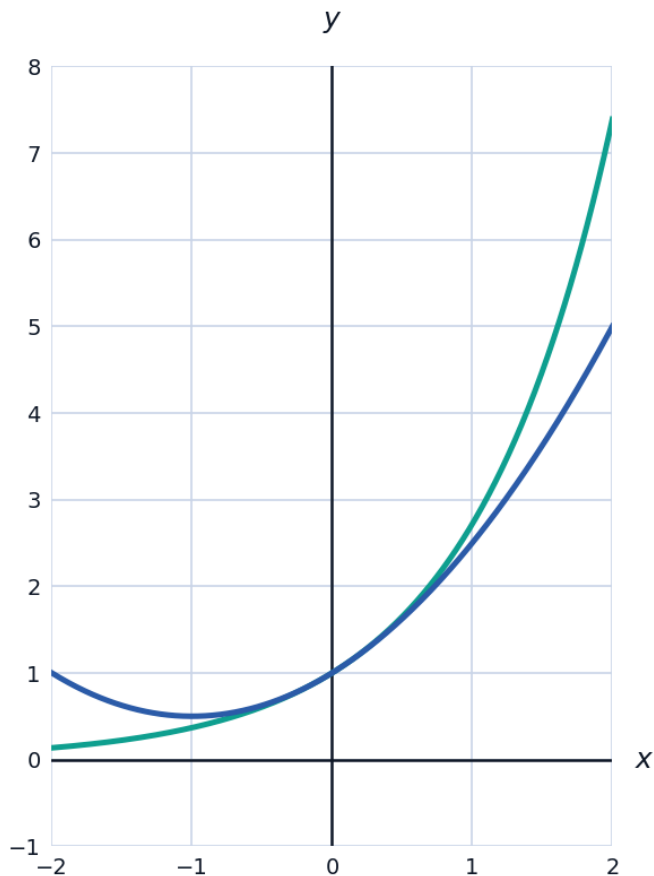
- 9 State the radius of convergence of the Maclaurin series for e^x , $\sin x$, and $\cos x$.
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Visualizing Taylor polynomial approximations

- 10 Compare $f(x) = \sin x$ to its degree-5 Maclaurin polynomial $P_5(x) = x - \frac{x^3}{6} + \frac{x^5}{120}$ on $[-\pi, \pi]$, and describe where the approximation is best and worst.



- 11 Compare $f(x) = e^x$ to its degree-2 Maclaurin polynomial $P_2(x) = 1 + x + \frac{x^2}{2}$ on $[-2, 2]$, and identify visually where P_2 underestimates or overestimates e^x .



Building Taylor series at a nonzero center

- 12 Find the Taylor series for $f(x) = \cos x$ centered at $a = \frac{\pi}{2}$, up through the cubic term.

Applying the Lagrange Error Bound to bound accuracy requirements

- 13 Using the degree-4 Maclaurin polynomial for $\cos x$ and the bound $|f^{(5)}(z)| \leq 1$, find the maximum error in approximating $\cos(0.3)$.
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The binomial series

- 14 State the binomial series for $(1+x)^k$ up through the quadratic term, and use it to find the Maclaurin series for $\sqrt{1+x}$ (i.e. $k = \frac{1}{2}$) up through the quadratic term.
- 15 Use the series found above to approximate $\sqrt{1.1}$ using the first three terms.
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Deriving new Maclaurin series from known ones

- 16 Use the known series for e^x to find the Maclaurin series for $\cosh x = \frac{e^x + e^{-x}}{2}$, up through the quartic term.